Numerical and Experimental Analysis of the Temperature Distribution in a Hydrogen Fuelled Combustor for a 10 MW Gas Turbine

Massimo Masi
e-mail: massimo.masi@unipd.it

Paolo Gobbato

Andrea Toffolo

Andrea Lazzaretto

Department of Mechanical Engineering,
University of Padova,
via Venezia, 1-35131 Padova, Italy

Stefano Cocchi
GE Oil & Gas—Nuovo Pignone SpA,
via Felice Matteucci, 2-50127 Firenze, Italy
e-mail: stefano.cocchi@ge.com

Proper cooling of the hot components and an optimal temperature distribution at the turbine inlet are fundamental targets for gas turbine combustors. In particular, the temperature distribution at the combustor discharge is a critical issue for the durability of the turbine blades and the high performance of the engine. At present, CFD is a widely used tool to simulate the reacting flow inside gas turbine combustors. This paper presents a numerical analysis of a single can type combustor designed to be fed both with hydrogen and natural gas. The combustor also features a steam injection system to restrain the NO_x *pollutants. The simulations were carried out to quantify the effect of fuel type and steam injection on the temperature field. The CFD model employs a computationally low cost approach, thus the physical domain is meshed with a coarse grid. A full-scale test campaign was performed on the combustor: temperatures at the liner wall and the combustor outlet were acquired at different operating conditions. These experimental data, which are discussed, were used to evaluate the capability of the present CFD model to predict temperature values for combustor operation with different fuels and steam to fuel ratios.* [DOI: 10.1115/1.4002017]

1 Introduction

A gas turbine combustor must satisfy a wide range of requirements such as high combustion efficiency, freedom from flame extinction, blowout, and pressure oscillations, low pressure losses and proper temperature distribution. In particular, the latter is a critical item for the durability of the hot gas turbine components. Combustion chamber wall overheating and hot spots at the combustor discharge must be avoided in order to maximize the operational life of the combustor, nozzle vanes, and turbine blades [1].

Usually, the temperature distribution at the combustor outlet is measured directly in experimental tests. The characterization of the thermal load on the liner walls is a bit complex due to the difficulties in carrying out measurements within the combustor [2]. References on liner wall temperatures are reported in Ref. [3].

The development of computational fluid dynamics (CFD) codes over the past few decades has provided a useful tool for estimating the temperature field within the combustor including the temperature distribution on the combustion chamber walls and at the turbine inlet as well [4–13]. Thus, CFD analyses, validated by reliable experimental data, have been used to investigate the liner wall thermal load [14,15] and to optimize the combustor outlet temperature field [16–18]. This approach can result in a significant reduction in the amount of experimental testing required since design modifications or new design solutions can be evaluated in the early stages of combustor development.

Many combustion models have been implemented into finite volume codes due to the complexity of the reacting flow physics. In particular, it is well-known that the temperature field within the combustion chamber is determined by the interaction between turbulence, chemistry, and heat transfer. Even though large eddy simulation (LES) has become an affordable tool for modeling flame/turbulence interactions [9,10], the traditional Reynolds averaged Navier–Stokes (RANS) approach is still the most widespread tool for gas turbine combustor modeling because it is an optimal compromise between accuracy and computational cost [19].

This paper presents a computationally low cost CFD analysis of the reacting flow inside a single can combustor. The calculations are aimed to compute the thermal field at different operating conditions and are carried out with the finite volume commercial code FLUENT® ver.6.2. The numerical model is specifically conceived for a future investigation of the thermoacoustic behavior of the combustor, which requires time-dependent analyses. Thus, a coarse grid and simplified numerical models (i.e., a hybrid global reaction mechanism and a simple two-equation turbulence model [4,8,11]) are adopted in order to reduce the computational time. The 3D geometrical model extends from the compressor discharge to the gas turbine inlet, coupling the reacting region of the combustor with the annulus "cold" flow domain. The advantages of this type of "coupled" analysis are reported in Ref. [16]. Some minor geometrical features are neglected so that a rough grid approach can be employed [20].

In the numerical simulations presented in this paper both natural gas and hydrogen full load operation are considered with or without steam injection ("wet" or "dry" operation). Thus the influence of the fuel type and steam dilution on the temperature field can be investigated. The computed temperature distribution along the liner wall and at the combustor outlet is compared with the experimental values measured during a full-scale test campaign. The testing activities were performed at ENEL's Sesta facility as part of a ministerial project coordinated by ENEL Ricerca (ENEL is the former national power company). The goal of the project was the design of a 10 MW class combined cycle gas turbine (CCGT) pilot plant to be fuelled with 100% hydrogen. The project was promoted by Hydrogen Park and was partly funded by Re-

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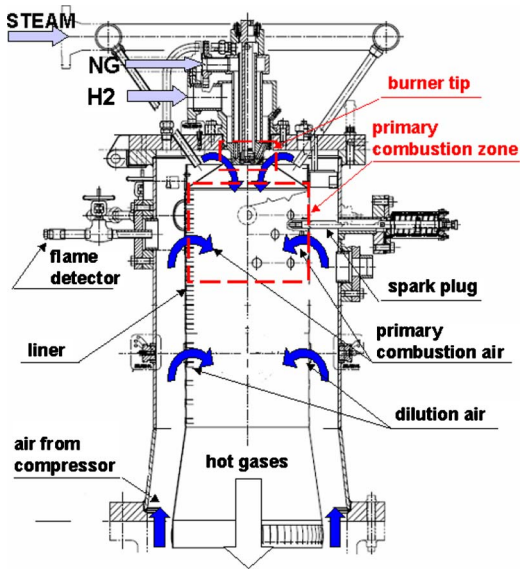

Fig. 1 Sectional view of the GE10 gas turbine combustor

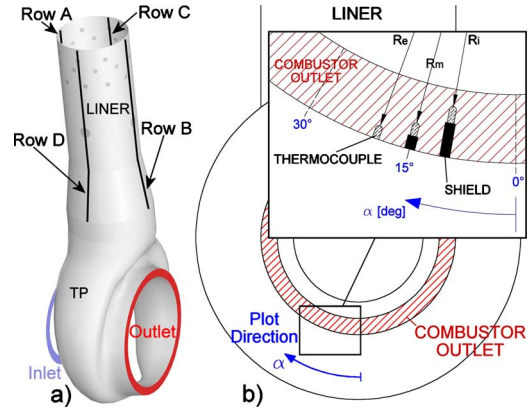

Fig. 2 (*a*) Rendering showing the rows of thermocouples installed to measure the liner wall temperatures. (*b*) Detail of the thermocouples installed in a 30 deg sector of the combustor outlet section.

gione Veneto; it was developed from 2006 to 2009 and involved six academic research partners [21–23]. A major research effort was focused on the development of the combustor to reduce NO_x emissions. In particular, two sets of combustor prototypes were designed over a 3 years period. All the prototypes were studied by means of full-scale test campaigns and CFD analyses. The first set of prototypes (the "base" prototypes) was obtained by coupling three different burners and two different liners. These prototypes, developed by GE Oil & Gas for the diffusive combustion of hydrogen, were tested and analyzed from 2007 to 2008. The results of these activities are reported in Refs. [20–22,24,25]. At the end of this first stage of the project, the prototype that showed the best performance was selected for installation in the new 10 MW class CCGT pilot plant at Fusina (near Venice) [21,23]. This combined cycle power plant is built within an existing ENEL coal fired power plant and can be fed with fuel blends ranging from pure natural gas to pure hydrogen.

In order to reduce the emission of NO_x, a second set of "enhanced" prototypes were conceived during the second stage of the project. The partnership between GE Oil & Gas, ENEL Ricerca and the other academic institutions resulted in combustor prototypes optimized for the injection of fuel and steam, which were tested in the summer of 2009 [23].

The combustor layout discussed in the following refers to one of the base prototypes, which was tested once again during the 2009 test campaign. This paper presents a comparison between the calculated profiles and measured values of temperature along the liner wall and the combustor outlet. Despite the usual practice in literature [15], the measured liner wall thermal loads are explicitly shown and discussed. All the measured temperatures are obtained from full load, full-pressure tests performed in operation with both pure natural gas and pure hydrogen.

2 The GE10 Combustor

The modeled combustor (Fig. 1) is the diffusive flame GE10-1 gas turbine combustor. It is a reverse flow single can combustor designed for diffusive flame, dual gas operation, and features steam injection for NO_x reduction. The compressor air stream is collected into a plenum (not represented in the figure) before entering the "annulus" (i.e., the annular enclosure between the liner and the casing). Then the air flows into the combustion chamber by means of the burner axial swirler, the primary and dilution holes and the liner cooling holes (grouped in several rows located along the entire length of the combustor). Natural gas and hydrogen are separately injected into the swirler slots without significant air premixing. The hot burnt stream reaches the first gas turbine stage trough a "transition piece" (TP) placed inside the air plenum.

Dilution steam flow is injected into the outer zone of the combustor head by a constant pitch series of azimuthal manifolds connected to a toroidal rail that surrounds the top of the casing.

3 Test Rig Configuration and Instrumentation

The combustion test rig is located at the ENEL experimental facility in Sesta (Tuscany, Italy). This facility is suitable for full-scale, full-pressure testing of a wide variety of combustion systems thanks to its very flexible multifuel supply systems. A train of two compressors assures a maximum air flow rate of up to 40 kg/s, which is employed both for combustion and rig cooling. Variable combustion air temperatures are achievable by means of electric heaters. During the GE10 combustor testing, the rig operates at its full capacity (1100 kg/h for hydrogen and 2500 kg/h for natural gas). The combustion air supply system is integrated with a special GE10 rig-adapter assembly, designed to reproduce on-engine compressor discharge and turbine inlet operating conditions. The GE10 gas turbine configuration is such that the combustor axis is perpendicular to the engine axis: hot gases exiting the annular section of the combustor are directed to the first stage turbine nozzles arranged in a proxy configuration having the same flow area of the on-engine component. Hot gases exiting the first stage proxy nozzle are then discharged directly into the exhaust duct.

The combustion test rig is equipped with resident instrumentation dedicated to the control of the air and fuel systems and which are necessary to ensure proper test rig operation. The mass flow rates are measured by standard orifices (for compressed air and steam) and flow meters (for natural gas and hydrogen). Pressure and temperature transmitters provide continuous monitoring of the air and fuel supply conditions. The liner is instrumented with several thermocouples, which give a discrete map of the metal temperature profile along four different rows parallel to the combustor axis (Fig. 2(*a*)). To measure the hot stream temperature, another three circular rows of thermocouples are installed at three different radii on the transition piece outlet. The azimuthal pitch of the thermocouples in each row is 30 deg (Fig. 2(*b*)).

The instrumentation also includes six dynamic pressure probes, a commercial flame detector and a video acquisition system. Pollutant emissions are measured by gas sampling at two different sections along the exhaust path by means of both chemiluminescence and infrared analysers. More rig features are available in Ref. [23].

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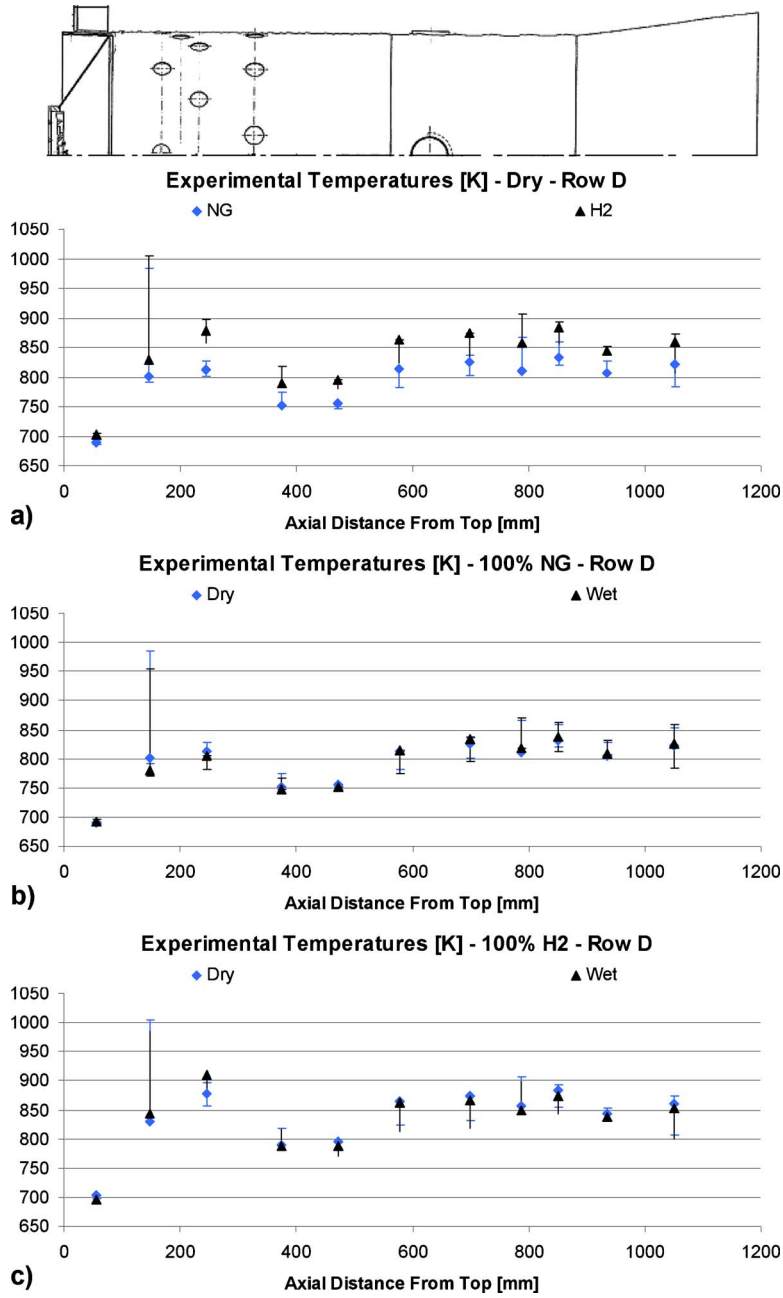

Fig. 3 Temperatures measured along the liner wall

4 Experimental Results

The results presented deal with the full speed full load operation of the engine since this is the condition associated with the highest thermal power and temperatures. The test conditions of the four experimental tests considered in this paper (pure natural gas feed and pure hydrogen feed both with and without steam injection) were derived from the GE10 gas turbine cycle simulator, a GE Oil & Gas proprietary tool. The simulator gives the test condition parameters (mass flow rates and physical properties of the inlet fluxes) for dry combustion of pure methane. This test condition is assumed as the reference for all other engine operations using different fuels (pure natural gas, pure hydrogen or blends of the two) and/or steam to fuel (SF) mass ratio, the operation parameters of which are derived by keeping the total thermal input unchanged. Thus, the SF ratio always refers to pure methane combustion, and for hydrogen or natural gas (or blends) operation the SF value is related to the amount of steam that would have been used to obtain the same heat release as that of the reference test condition. In the wet cases, only the SF=1 operating condition is considered.

Figure 3 shows the liner metal temperatures against the axial coordinate. The markers refer to the data about row D. The values of the other rows are presented as error bars (each bar ranges between the maximum and the minimum temperature measured in the four rows for the same axial position).

All the curves exhibit a temperature peak in the primary zone and a further temperature increasing downstream of the dilution holes. The temperature spread at the same axial position denotes that the reacting flow is not perfectly axisymmetric. This is mainly due to the irregular pattern of the primary holes, which causes a different interaction between the swirler spot jets and the primary air jets. However, the values never reach warning limits. It is not

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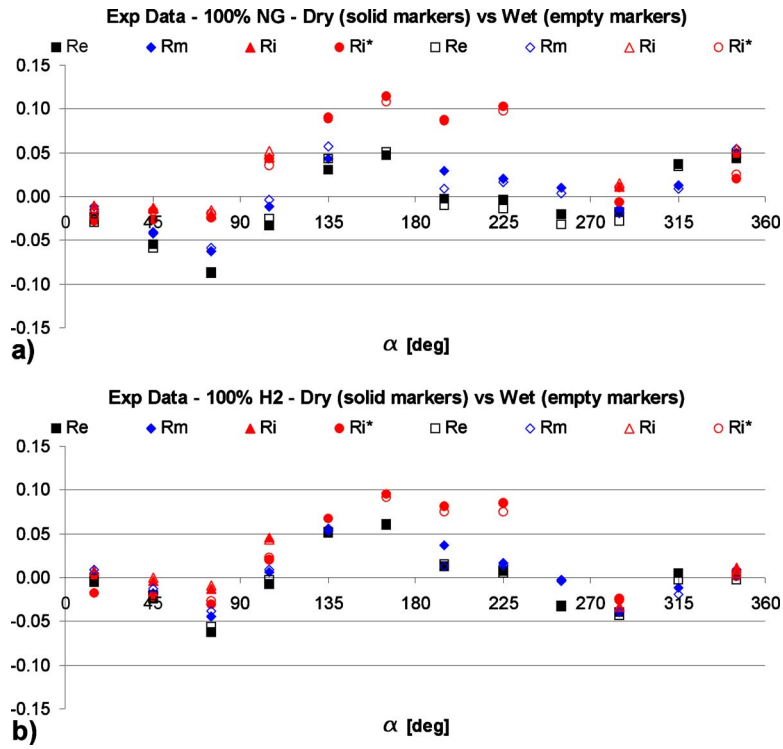

Fig. 4 Nondimensional temperatures along the three circular rows indicated in Fig. 2(*b*)

assured that the spread in liner temperatures observed in the primary zone (about 200°C) depends on the malfunction of a thermocouple.

The temperatures for 100% natural gas and 100% hydrogen are similar (Fig. 3(*a*)). The small differences are probably due to the different flame shapes (combined effect of fuel jet's penetration and reactivity) and/or slightly different operating conditions. The influence of steam on the metal temperature is low and it is restricted mainly to the primary zone (Figs. 3(*b*) and 3(*c*)).

Figure 4 shows the measured temperatures at the combustor outlet against the angular coordinate. The data are made nondimensional using the average value of the measurements in order to emphasize both the radial and the azimuthal nonuniformities in the temperature distribution. Since half of the thermocouples in the inner row (red triangular markers) were out of order during the test, one additional test is included in the graphs to provide the missing values (red circular markers). This test (identified by an asterisk) was performed under very similar combustor operating conditions some weeks before the tests considered in this paper.

Both diagrams show a slightly nonuniform temperature distribution. In particular, a temperature peak appears at the top side of the discharge (at about 180 deg) and two local minima occur near 90 deg and 270 deg. Moreover, the data show a temperature increase going from the outer row to the inner row of thermocouples.

By comparison with Figs. 4(*a*) and 4(*b*), it appears that shifting from natural gas to hydrogen operation does not change the data layout significantly. Furthermore, the temperature distribution at the combustor outlet is not influenced substantially by steam injection, as previously stated for the liner (Fig. 3).

5 Computational Model

The geometrical model employed in the simulations is represented in Fig. 5(*a*). The model includes the annulus, the burner, the liner, the air plenum, and the TP. The overall length of the computational domain is about 2700 mm.

In order to use a rough grid approach, some minor geometrical features were simplified or neglected. In particular the liner cooling holes and the steam manifolds were not included in the geometrical model. However, their function is preserved by means of a porous wall model, which assures the flow of the cooling air stream through the liner. A similar approach is used in Refs. [18,24]. The remaining geometrical details (liner wall thickness,

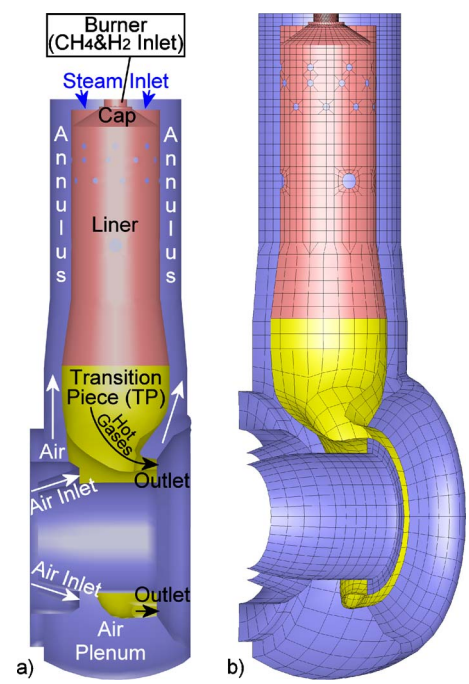

Fig. 5 (*a*) Section of the geometrical model. (*b*) Section of the computational grid

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axial swirler slots, fuel injection channels, plenum and transition piece shape) are modeled without simplifications.

An almost fully hexahedral coarse grid of about 55,000 cells was set up (Fig. 5(*b*)). Tetrahedral, pyramidal, and triangular prism cell types were employed at the burner outlet and in the zone that links the liner to the transition piece. In addition to the already mentioned geometrical modeling constraints, the rough grid approach forces the near wall refinement to be neglected. Despite the inaccuracies expected in the flow field (the boundary layer viscous stresses are not accurately computed), the air splitting percentages between the several liner and burner passages are guaranteed by the proper setup of the porous media inertial resistance coefficients (see the discussion in the following sections). Furthermore, the coarseness of the grid does not allow the detailed thermal flow field inside the combustor to be captured. In fact, the grid size might describe only the main effects of turbulence and chemistry (especially in the primary combustion zone). This results in a temperature field roughly equivalent to that of the real one. However, in the opinion of the authors, the approximation level due to this low cost computational approach is consistent with the aim of the nonstationary analyses planned for this model.

The fluid flow is modeled as a compressible ideal gas mixture, thus the physical properties are calculated as a weighted sum of the mass fraction of the mixture components. The specific heat of each gas is computed using a fourth order polynomial law [26].

The combustion is modeled by means of the hybrid finite-rate/eddy-dissipation model considering a one-step reaction both for the methane[1] and the hydrogen.

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

$$2H_2 + O_2 \rightarrow 2H_2O$$

In this hybrid model both the Arrhenius and eddy-dissipation reaction rates are calculated and the net reaction rate is the lower of these two rates. Therefore, the Arrhenius model controls the ignition and the kinetics of the reaction and the flame is controlled by the turbulent mixing through the eddy-dissipation model [27]. In the Arrhenius model the values of the pre-exponential factor, the rate exponents and the activation energy are specified as suggested in Refs. [28,29]. The temperature exponent is assumed to be zero both for the methane and the hydrogen. The eddy-dissipation coefficients are set equal to the standard values (4 and 0.5) [30]. The combustion model requires the solution of three other transport equations for the mass fraction of hydrogen or methane, oxygen, carbon dioxide, and water (the nitrogen mass fraction is derived algebraically). The molecular diffusivity required by each of these transport equations is calculated using a simplified model based on the gas kinetic theory [31].

The turbulence is modeled by means of the k-ω SST model [32], also successfully employed in Ref. [11], and the standard wall functions for momentum and energy are used. The computationally low cost P-1 model [33] is chosen for radiation heat transfer. The absorption coefficient of the mixture is computed by the weighted sum of the gray gases approach [34].

The four experimental tests already discussed provide the set of boundary conditions for the simulations. A mass flow rate condition is specified at the plenum inlet section, at the inlet of the fuel channels and at the steam inlet ports, respectively. The temperature, the pressure and the composition of all the fluxes are known from measurements. A constant pressure condition is imposed at the combustor outlet. The outer boundaries of the model (casing, plenum, fuel channels, and swirler walls) are considered to be adiabatic (this assumption is justified by the experimental experience). Instead, heat transfer occurs through the transition piece wall, so the air flowing into the plenum is warmed by the inner hot gas stream.

[1]The simplified global reaction mechanism requires the natural gas to be considered as pure methane. The difference in LHV is taken into account in the simulations to keep the thermal input unchanged.

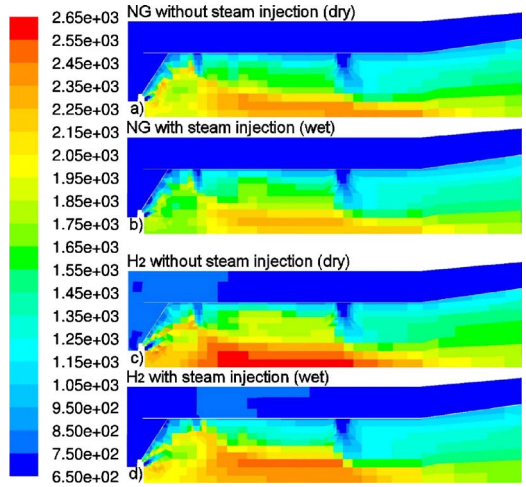

Fig. 6 Temperature fields around the liner [K]

Both the liner and cap louvered walls are modeled by means of an isotropic porous media to take into account the metal cooling. In order to correctly estimate the heat exchange between the inner hot stream and the annulus air flow, a uniform porosity is specified in the media, which is calculated according to liner design specifications. In the solid fraction of the media the conduction heat transfer is modeled. The inertial resistance coefficients of the various surface patches are tuned to fit the airflow distribution data (supplied by ENEL) along the liner derived by an experimental campaign on an isothermal flow test rig.

The unsteady RANS segregated solver with the SIMPLE algorithm [35] is used. Convergence is reached by means of a time-marching approach with a second order implicit time discretization scheme (time step equal to 100 μs). For each simulation, the solution is judged to be converged when the mass flow rate and the temperature at the combustor outlet do not show any appreciable variation in time. On the average, convergence is reached after 10,000 time steps (1 s). The CPU time needed to perform 0.1 s of simulation is about 6 h with an AMD Athlon® 4600+ Dual Core (2.4 GHz) processor. For more details on the physical models and numerical simulation setup, see Ref. [25].

6 Numerical Results

Figure 6 shows the calculated temperature scalar field around the liner. The contour plot section plane encloses thermocouple rows C and D (see Fig. 2(*a*)). The temperature fields help in understanding how the development of the reacting flow influences the liner wall temperatures. Starting from the top of the liner, the wall temperature increases abruptly before the primary air holes due to the ignition of the swirling mixture that licks the liner inner surface. In the primary zone, the air jets drive the hot reacting stream around the combustor axis and the wall temperatures decrease. The hot axial flow is then partially scattered by the deep penetration of the dilution jets. The temperature increase downstream of the dilution holes is caused by the spreading of the hot axial stream and the lower amount of cooling air delivered to this portion of the liner compared with the primary combustion section. The temperature maxima are reached with hydrogen operation due to the higher flame temperature compared with that of natural gas. Steam injection (Figs. 6(*b*) and 6(*d*)) does not significantly modify the shape of the reacting flow within the fields. However, the reduction in the temperatures both in the ignition zone and in the axial stream leaving the primary combustion zone is clear.

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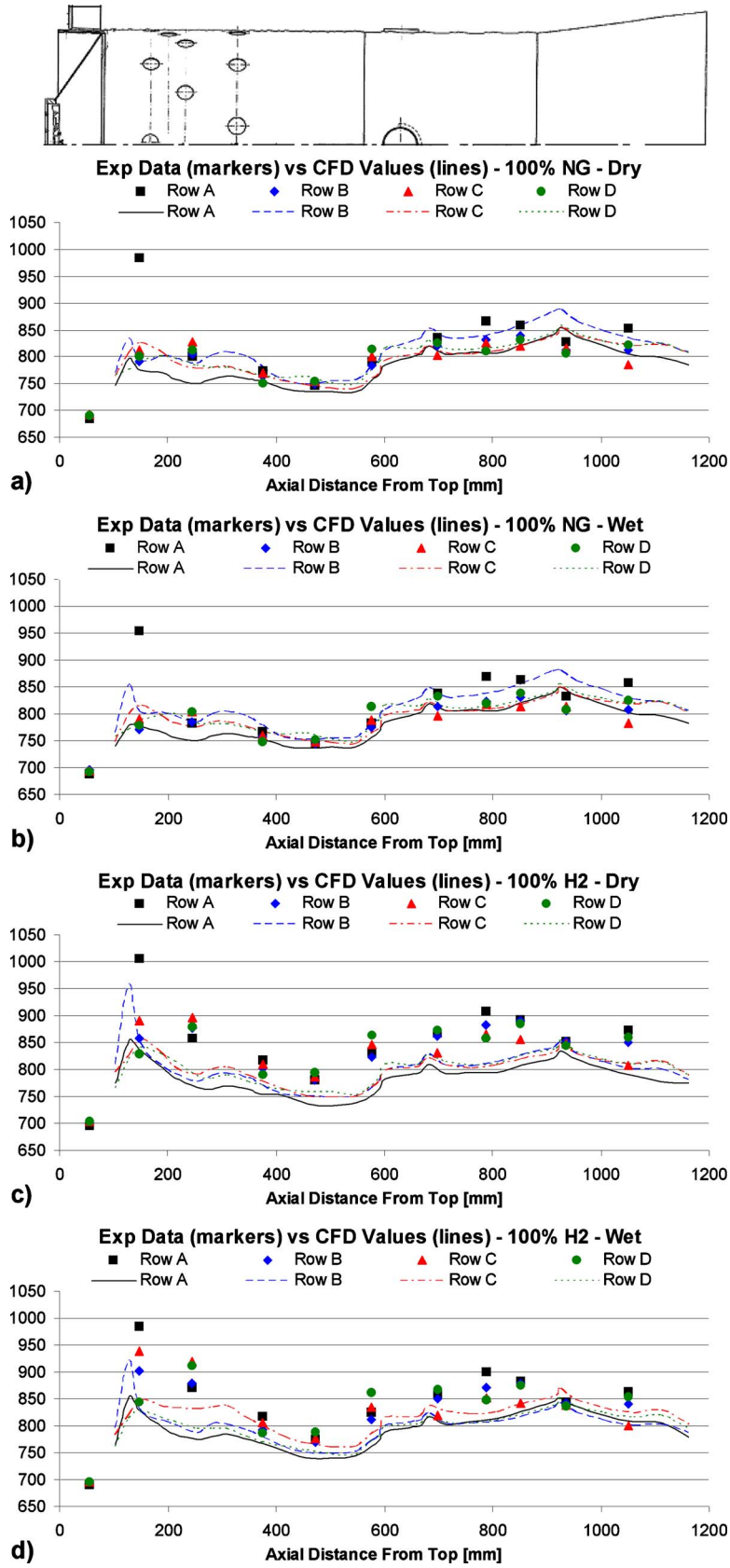

Fig. 7 Comparison between the experimental data and temperature profiles along the liner wall (see Fig. 2(*a*) for the arrangement of the rows) [K]

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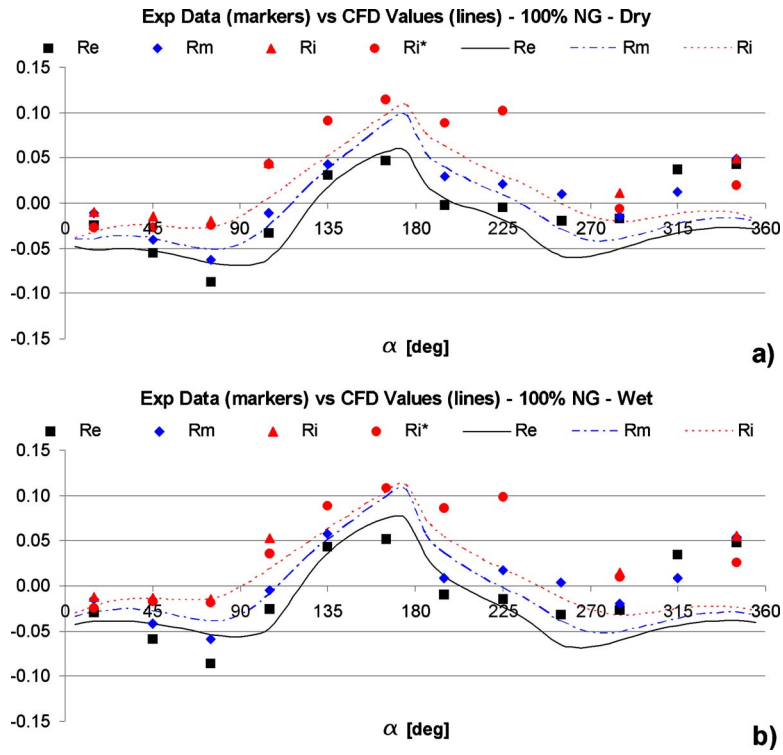

Fig. 8 Natural gas: comparison between the experimental data and temperature profiles along the circular rows shown in Fig. 2(*b*)

Figure 7 compares the temperature distribution along the four thermocouple rows (A–D) obtained from the CFD calculation (lines) and the measurements (markers). The experimental values are quite well fit by the numerical profiles in natural gas operation (see diagrams 7(*a*) and 7(*b*)). Instead, the calculated profiles are on the average lower than the measured data when hydrogen is used (diagrams 7(*c*) and 7(*d*)). The comparison also shows a difference in the axial coordinate of the temperature peaks, which is mainly due to the grid coarseness. Moreover, the measured maximum in row A is not completely captured by the calculated results

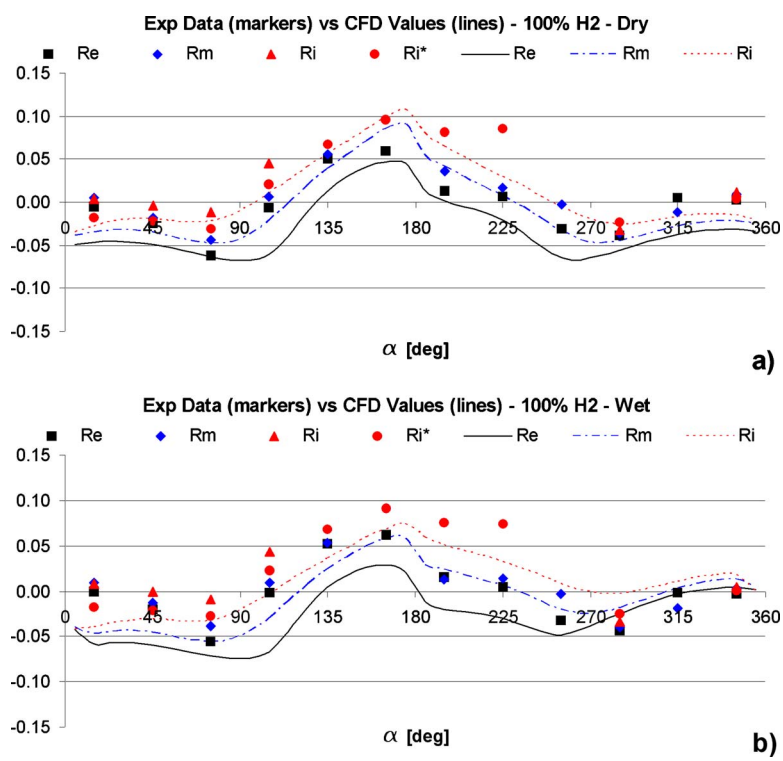

Fig. 9 Hydrogen: comparison between the experimental data and temperature profiles along the circular rows shown in Fig. 2(*b*)

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for profile B. Further analysis must be performed to better understand the causes of this peak value of the liner wall temperature.

Figures 8 and 9 present the CFD temperatures at the combustor outlet. The computed values are made nondimensional using the same procedure already employed in Fig. 4. The graphs show good agreement between the measured data and numerical profiles both for natural gas (Figs. 8(*a*) and 8(*b*)) and hydrogen (Figs. 9(*a*) and 9(*b*)) operation. The difference between the average of the experimental and the calculated temperatures is 10°C for natural gas dry operation. The inspection of the temperature field inside the TP suggests that the hot zone in the upper part of the combustor discharge is due to the high temperature axial stream leaving the core of the liner.

7 Conclusions

This paper deals with the numerical and experimental analysis of the temperature distribution in a hydrogen fuelled combustor for a 10 MW gas turbine. Both natural gas and hydrogen full load operating conditions with and without steam injection (wet or dry operation) were studied. The computational domain was a 3D geometrical model extending from the compressor discharge to the gas turbine inlet. A computationally low cost approach was employed in view of using the model in time-dependent calculations.

A full-scale test campaign was performed on the combustor: temperatures at the liner wall and the combustor outlet were acquired at many operating conditions. Four sets of experimental data were employed for comparison with the CFD results.

The experimental values measured by four rows of thermocouples along the liner wall showed that the distribution of the metal thermal load was not perfectly axisymmetric. At the combustor outlet, a slightly nonuniform distribution of temperatures also occurred.

The thermal fields obtained from the simulations clarified the influence of the fuel type and steam injection on the temperature distribution. Despite the low accuracy expected from a coarse grid calculation, the computed temperature profiles showed acceptable matching with measurements in the liner wall. Moreover, good agreement was found between the CFD profiles and experimental data at the combustor discharge at all the operating conditions considered.

The agreement with the measured data suggests that the computationally low cost CFD model presented is able to capture the mean temperature field within the combustor, which is the fundamental requirement for the planned thermoacoustic characterization of the analyzed combustor.

Nomenclature

CCGT = combined cycle gas turbine
dry = without steam injection
NG = natural gas
R_i, R_m, R_e = inner, medium and outer radius of the outlet thermocouple rows
SF = steam to fuel mass ratio
SST = shear stress transport
TP = transition piece
wet = with steam injection

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